Poushali Deb Purkayastha

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EDUCATION

Master of Science in Applied Data Intelligence, San Jose State University, CA — **Aug 25 – May 27**
Relevant Courses: Data Warehousing and Pipeline Development, Data Visualization, Distributed Systems, Machine Learning, Deep Learning.

Bachelor's in computer science & engineering, SRM Institute of Science & Technology, India — **Jun 18 – May 22**
Relevant Courses: Data Structures and Algorithms, Operating Systems, Network Security, Data Mining, DBMS, and Advanced Programming.

SKILLS & INTERESTS

Agentic AI: LLM Orchestration, RAG Pipelines, Multi-Agent Systems, MCP Protocol, Prompt Engineering, LangChain, LangGraph
LLMs & ML: Claude, ChatGPT, Gemini, Redis Embeddings, TensorFlow, PyTorch, Anomaly Detection
Data Engineering & Pipelines: ETL/ELT, Apache Spark, PySpark, Airflow, Kafka, Telemetry Pipelines, Apache Iceberg, Data Fabric
Cloud & Infrastructure: AWS (S3, EC2), Docker, Kubernetes, CI/CD, MinIO, Databricks (familiar)
Databases: Snowflake, BigQuery, PostgreSQL, MySQL, Redis
Programming: Python, SQL, Java, Go (basic)

WORK EXPERIENCE & INTERNSHIPS

Data Engineer | Deloitte — **Jan 23 – Jul 25**

- Engineered complex ML systems via Agentic AI-powered ETL pipelines with RAG-based validation and AI guardrail techniques across large-scale distributed datasets, including LLM evaluation, fine-tuning, and knowledge distillation cycles to validate and optimize pipeline outputs against production baselines, reducing validation failures 30% and contributing to $75K+ in operational savings.
- Optimized Snowflake data models for high-volume retrieval workflows, cutting query execution time 35% while balancing engineering rigor with high-performance computing demands.
- Deployed on-demand containerized microservices (Docker, Kubernetes) with CI/CD across distributed systems, reducing production incidents 40%, balancing research exploration with engineering rigor and operational reliability at scale.

Backend & Data Engineer | HighRadius Technologies — **Jun 21 – Jul 22**

- Reduced manual data processing effort by 50% by developing automated backend pipelines supporting large-scale, AI-driven data systems, operationally reliable across high-volume structured datasets.
- Improved data quality by 40% by redesigning relational database schemas and deploying automated validation pipelines, ensuring reliable data feeds for downstream ML models.
- Increased system throughput by 30% by optimizing distributed processing workflows and multithreaded backend services within a large-scale distributed systems environment.

PROJECTS

Text-to-Pipeline AI Agent | LLM-Driven OSS Application — **Apr 26 – May 26**

- Architected a natural language to live pipeline agent using GPT-4o and LangChain on the RocketRide OSS engine, enabling any user to generate and deploy a fully functional AI pipeline from a single English sentence with zero manual JSON authoring.
- Built end-to-end execution pipeline with real-time visual DAG animation in VS Code, Input → LLM → Output, with automated credential injection, unique project GUID generation, and dual-interface support via Streamlit UI and Python CLI.

Chameleon of the Dungeons | LLM-Driven Agentic Honeypot System — **Jan 26 – Feb 26**

- Built a self-evolving complex ML system with a synthetic data pipeline autonomously generating operational rules in real time via FastAPI agent framework, Redis-backed RAG pipeline, and streaming data transformations.
- Engineered a multi-agent LLM system with multi-model orchestration over transformer-based open-source models (Llama, GPT-4) and public APIs, applying knowledge distillation and student-teacher principles to optimize model selection for latency-sensitive inference, NLP techniques for real-time threat intelligence extraction, and AI security validation — top-3 finish among 60+ teams at WeaveHacks 3.

Serverless Data Lakehouse | Medallion Architecture — **Jan 26 – Apr 26**

- Architected a Bronze-Silver-Gold Medallion Lakehouse on on-demand, containerized AWS S3-compatible infrastructure, a high-performance computing environment with zero-copy agentic data federation principles built for infrastructure-heavy ML projects.
- Engineered end-to-end PySpark synthetic data pipeline ingesting, cleansing, and transforming 3M+ raw geospatial records into Gold-layer analytics tables; implemented Apache Iceberg for ACID transactions, schema evolution, and Time Travel queries — pipeline architecture designed to support downstream ML training workflows, including transformer and CNN-based model evaluation.

Distributed Media Processing Pipeline — **Sep 25 – Oct 25**

- Built a large-scale distributed system (FastAPI, Celery, Redis, Kubernetes) with autoscaling and queue-based workload distribution, production-grade orchestration demonstrating high-performance computing principles and operational reliability across burst-load scenarios.